

Week 8 : Assignment 8

Your last recorded submission was on 2026-09-13, 16:33 IST.

1. Consider the following 2 commands in a system that supports discretionary access control.

```
command CONFER_execute(S, S', O)
    If o in A[S, O] then
        Enter x in A[S', O]
    End
```

```
command ADD_write(S,O)
    If x in A[S,O] then
        enter w in A[S,O]
    End
```

Which one of the following statements is TRUE?

- A. There is no leakage of rights due to these commands
- B. A leakage may occur when an owner of an object executes CONFER_execute() command.
- C. A leakage may occur when a user with execute permissions, executes ADD_write() command

- ☐ Only A is correct
- ☐ Both B and C is correct
- ☒ Only B is correct
- ☐ Only C is correct

2. Match the following

- a) Trojan-horse 1) excess data corrupts values in adjacent memory addresses
- b) Buffer Overflow 2) memory is freed more than once for a specific memory address
- c) Double frees 3) malicious code disguising as legitimate code to claim it's privileges

- ☐ a-1, b-3, c-2
- ☐ a-2, b-1, c-3
- ☐ a-1, b-2, c-3
- ☒ a-3, b-1 c-2

3. State True/False:

In the Biba model, information from a top secret object will not flow to secret clearance.

- ☐ True
- ☒ False

4. State True/False

In the Bell-Lapadula model, information from a top secret object will not flow to secret clearance.

- ☒ True
- ☐ False

5. A user X with clearance "Secret" decides to transfer information to a third party Y and Y tries to make changes in the "Top-Secret" class. This breach of information can be prevented by _____.

- ☐ Implementing access control matrix
- ☒ Implementing Bell-LaPadula model
- ☐ Implementing the Biba Model
- ☐ None of the above

6. Match the following

- | | |
|--------------|---|
| a) Gadget | 1) pseudo random number which detects buffer overflow |
| b) ASLR | 2) Short sequence of instructions ending in a return |
| c) Shellcode | 3) May execute in the stack if the NX bit is disabled |
| d) Canaries | 4) helps prevent ROP attacks |

- ☐ a-4, b-1, c-2, d-3
- ☒ a-2, b-4 c-3, d-1
- ☐ a-1, b-3, c-4, d-2
- ☐ a-2, b-3, c-4, d-1

7. Arrange the following, based on the order in which the following parameters are pushed onto the stack:

- a) Return address is pushed
- b) The function call parameters
- c) Local variables are allocated
- d) Previous frame pointers

- ☐ a, b, c, d
- ☐ b, c, d, a
- ☒ b, a, d, c
- ☐ a, b, d, c

8. To make sure that your source code has minimum number of buffer overflow vulnerabilities, which of the following would you need to look out for:

- ☐ Printf & scanf in the code
- ☐ gets() in the code
- ☐ Loops that manipulate arrays
- ☒ All of the above

9. An ROP attack can be potentially detected at run-time by:

- ☒ pseudo random number verification after start and before end of functions
- ☐ The number of return statements present in the executable
- ☐ The number of return statements executed in the processor
- ☐ By a non-executable stack

10. State True/ False:

ASLR randomizes the position of all variables in the stack and thereby making it difficult to perform buffer overflows.

- ☐ True
- ☒ False